

Bilirubin, Total

Test Details

METHODOLOGY

Colorimetric

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Causes of high bilirubin: Liver disease: hepatitis, cholangitis, cirrhosis, other types of liver disease (including primary or secondary neoplasia); alcoholism (usually with high AST (SGOT), GGT, MCV, or some combination of these findings); biliary obstruction (intrahepatic or extrahepatic); infectious mononucleosis (look also for increased LD (LDH), lymphocytosis); Dubin-Johnson syndrome; Gilbert disease¹ (familial hyperbilirubinemia) is encountered as a moderate elevation with otherwise unremarkable chemistries.

Anorexia or prolonged fasting: 36 hours or more may cause moderate rise.

Pernicious anemia, hemolytic anemias, erythroblastosis fetalis, other neonatal jaundice, hematoma, and following a blood transfusion, especially if several units are given in a short time.

Pulmonary embolism and/or infarct, congestive heart failure.

Drugs: A large number of drugs can cause jaundice by in vivo action or by chemistry methodology. Drugs causing cholestasis and/or hepatocellular damage include diphenylhydantoin, azathioprine, phenothiazines, erythromycin, penicillin, sulfonamides, oral contraceptives, anabolic-androgenic steroids, halothane, aminosalicic acid, isoniazid, methyldopa, indomethacin, pyrazinamide, and others.

Custom Additional Information

Interpretation of increased bilirubin is greatly enhanced by other chemistry results. In acute viral hepatitis with jaundice, for instance, the transaminases ALT (SGPT) and AST (SGOT) are consistently increased, while an isolated elevation of bilirubin is seen in Gilbert disease.¹

Obstruction causes increases in bilirubin and alkaline phosphatase greater than and out of proportion to the transaminases.² Amylase and lipase are useful in differential diagnosis of obstructive jaundice. In **intrahepatic cholestasis**, the transaminases are not as increased, relative to bilirubin, as they are in hepatitis.³ Work-up of jaundice has been outlined.^{4,5}

Nicotinic acid increases the formation of bilirubin in the spleen, leading to a rise in unconjugated bilirubin. This can be used as a test for Gilbert disease¹ in which there is a decreased hepatic clearance of unconjugated bilirubin. Although the indirect bilirubin level is increased in normal controls when nicotinic acid is given, the increase is much greater in patients with Gilbert disease. In the Crigler-Najjar syndrome type I, the unconjugated bilirubin is >20 µg/dL. In type II, the level is <20 µg/dL.

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	3 days
Refrigerated	3 days
Frozen	14 days
Freeze/thaw cycles	Stable x2

Reference Range

See table.

Age	Range (mg/dL)
Newborns, term, and near term	
24 hours	0.0–8.0
48 hours	0.0–13.2
72 hours	0.0–15.6
96 hours to 1 month	0.0–16.6
Children ≥1 month and adults	0.0–1.2

Storage Instructions

Room temperature

Causes for Rejection

Gross hemolysis; improper labeling; gross lipemia

Footnotes

1. Ohkubo H, Okuda K. The nicotinic acid test in constitutional conjugated hyperbilirubinemia and effects of steroids. *Hepatology*. 1984 Nov-Dec;

4(6):1206-1208. PubMed 6500512 (<http://www.ncbi.nlm.nih.gov/pubmed/6500512>)

2. Scharschmidt BF, Goldberg HI, Schmid R. Current concepts in diagnosis. Approach to the patient with cholestatic jaundice. *N Engl J Med*. 1983 Jun 23; 308(25):1515-1519. PubMed 6855824

(<http://www.ncbi.nlm.nih.gov/pubmed/6855824>)

3. Goldberg DM, Spooner RJ, Ellis G, Ward AM. Biochemical features of intrahepatic cholestasis. *Am J Clin Pathol*. 1979 May; 71(5):557-563.

PubMed 453073 (<http://www.ncbi.nlm.nih.gov/pubmed/453073>)

4. Ostrow JD. Jaundice in older children and adults. In: Lundberg GD, ed. *Using the Clinical Laboratory in Medical Decision Making*. Chicago, Ill:

American Society of Clinical Pathologists;1983:41-48.

5. Fischer MG, Gelb AM, Weingarten LA. Cholestatic jaundice in adults. In: Lundberg GD, ed. *Using the Clinical Laboratory in Medical Decision Making*. Chicago, Ill: American Society of Clinical Pathologists;1983:49-54.